



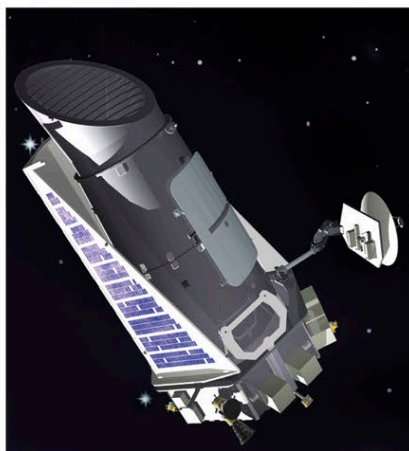
△ CoRoT

One of the main mission objectives of the CoRoT spacecraft, launched in 2007, was to detect transiting extrasolar planets. After finding 25 exoplanets, it was retired in 2013.

MULTIPLANETARY SYSTEMS

MANY EXOPLANETS RESIDE WITHIN MULTIPLANET SYSTEMS—GROUPS OF TWO OR MORE PLANETS ALL ORBITING THE SAME DISTANT STAR OR EVEN, IN SOME CASES, A PAIR OF STARS THAT ARE THEMSELVES CIRCLING EACH OTHER.

These intriguing multiplanet systems are quite diverse in terms of the mix of different sizes of planets they contain, the types of host star, and the number of planets that orbit within the host star's (or stars') habitable zone. Hundreds have been found already, at distances ranging from a few light-years to thousands of light-years from Earth. Only a few of the systems that have been discovered bear much resemblance to our Solar System, although a handful contain one or more roughly Earth-sized planets within a star's habitable zone, and so hold out the possibility of harboring life. But new systems are regularly detected, and data about the planets already found is frequently being updated, so this situation is constantly changing.

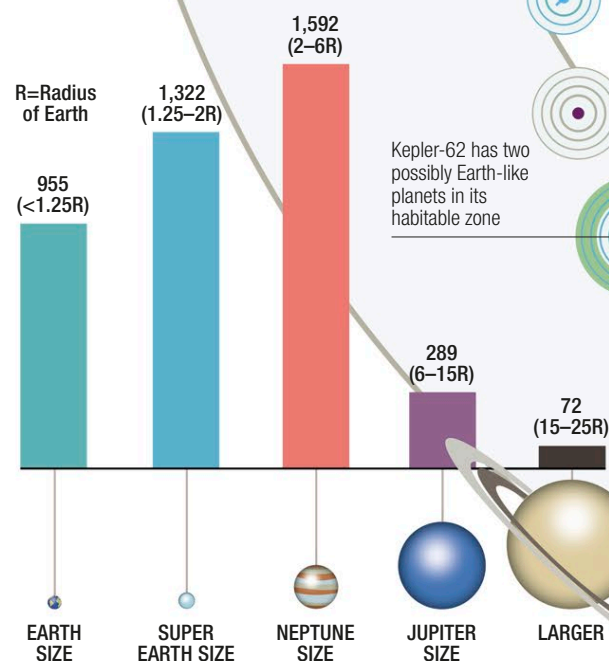


△ Kepler

Since its launch in 2009, NASA's Kepler space telescope has sought out exoplanets, particularly Earth-sized ones, using the transit method. By early 2016, it had detected 84 multiplanetary systems, each containing between two and seven planets, as well as as many single planet systems.

▷ Planet types

This bar graph shows the numbers of different sizes of all exoplanets (both confirmed and unconfirmed) up to early 2016. The different sizes are defined by radius in comparison with Earth's radius (so Super-Earths, for example have radii between 1.25 and 2 times Earth's radius).



24 Sextantis has two Jupiter-sized planets that dance around each other gravitationally

HIP 57274

HD 134606

Kepler-186 contains the first Earth-sized exoplanet found orbiting in a star's habitable zone

PSR 1257+12 is a pulsar with two super-Earths and one other tiny planet orbiting it

Kepler-62 has two possibly Earth-like planets in its habitable zone